

Quantum-Mechanics-1

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PY511

Contents

Lecture 1: Syllabus day

Tue 02 Sep 2025 13:38

Should have access to the server at `buphy.by.edu` and should be able to see 511. Check later.
QM is 100 years old this year woow

we just reviewed stern-gerlach. ai use allowed provided its declared and prompt is included, but like yknow

Ok we did do the axioms of QM.

- The (pure) physical state of a particle is completely described by an element called a *ray* in a complex Hilbert space $\mathcal{H}$.
- bruh he is reminding us what a hilbert space is

Lecture 2: Axioms of quantum mechanics

Thu 04 Sep 2025 13:57

The axioms of Quantum Mechanics are as follows.

1. (Dirac) The (pure) physical state of a particle is completely described by a vector in a complex Hilberts space $\mathcal{H}$ with the canonical norm.
2. (Schrödinger) The time evolution dynamics are governed by the Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\Psi\rangle = \hat{\mathbf{H}} |\Psi\rangle,$$

where $\hat{\mathbf{H}}$ is the Hamiltonian operator.

3. Consider a composite system of two different Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$. Given $|A\rangle \in \mathcal{H}_1$ and $|B\rangle \in \mathcal{H}_2$ the subsystems, the total physical system is completely described by the vector $|A\rangle \otimes |B\rangle$ in the space $\mathcal{H}_1 \otimes_{\mathbb{C}} \mathcal{H}_2$. We write $|A; B\rangle$ or $|AB\rangle$ for $|A\rangle \otimes |B\rangle$.
4. A physical observable is represented by a Hermitian operator $\hat{\mathbf{A}}$ on the elements of a Hilbert space such that the measurement on a system in a state $|\Psi\rangle$ always returns just one of its eigenvalues λ_n $|\psi_n\rangle = \hat{\mathbf{A}} |\psi_n\rangle$ with probability of measuring λ_n given by $\langle \psi_n | \Psi \rangle$.
5. (Copenhagen) After the measurement, if the measurement yields eigenvalue λ_n , then the system is collapsed to a corresponding eigenstate $|\psi_n\rangle$.
6. (Many Worlds, alternative to (5)) After the measurement, the system and observer become entangled. The system continues to evolve according to (2).

Remark 0.0.1. The Schrödinger equation can be fully relativistic, even though it appears to violate special relativity, so long as the Hamiltonian respects Lorentz invariance. However, the standard Schrödinger equation in the position basis

$$\hat{H} = \frac{\hat{P}^2}{2m} + \hat{V}(\hat{x}),$$

violates Lorentz invariance. Unfortunately, this can also only be done in flat spacetime. The equation is, however, fully deterministic.

Lecture 3: Single particle interferometers

Tue 09 Sep 2025 13:46

Particle sent through 50/50 beamsplitter. Both induced “beams” bounce off mirrors into a second beam splitter, recombining and sending to one of two mirrors. We input (α, β) , and we get an output (α', β') . We need a Hermitian operator. a aygugggg

Lecture 4: Solving the Schrodinger Equation

Thu 11 Sep 2025 13:50

Let $\mathcal{H}$ be a Hilbert space, and let $|\psi_0\rangle$ be some element in $\mathcal{H}$ which represents an initial state. There is then some operator mapping $|\psi_0\rangle \mapsto |\psi(t)\rangle$. We thus have an operator

$$|\Psi(t)\rangle = U(t, t_0)|\Psi(t_0)\rangle.$$

We thus want to solve for the time evolution operator $U(t, t_0)$. This should satisfy some “equation of motion” in $\mathcal{H}$ inherited from Schrödinger’s equation. This gives (writing H, U for the operators)

$$i\hbar \frac{\partial}{\partial t} U|\Psi_0\rangle = HU|\Psi_0\rangle$$

for all vectors. Since the vector is not time dependent, we have

$$i\hbar \left(\frac{\partial}{\partial t} U \right) |\Psi_0\rangle = HU|\Psi_0\rangle.$$

Since this must be true for *all* vectors, they cancel:

$$\frac{\partial U}{\partial t} = -\frac{i}{\hbar} HU.$$

The operator space is only a Banach space, since it has no inner product. This gives us an equation in an operator space. We solve this equation, subject to the obvious initial condition that $U(t_0, t_0) = 1$ (1 is the identity; U is operator-valued). We of course intuitively think that

$$U = C \exp \left(-\frac{i}{\hbar} H \right).$$

Let’s try to rigorize this intuition. Is U an observable? Well,

$$\frac{\partial U^\dagger}{\partial t} = \frac{i}{\hbar} U^\dagger H,$$

since H is an observable, where $U^\dagger(t_0, t_0) = 1$. These equations are different in general, so U is not an observable in general. We also know that $\langle \Psi | \Psi \rangle = 1$, so adding in U gives $U^\dagger U = 1$. Therefore, U is unitary, and thus preserves the inner product. We claim it follows that $U(t, t_0) = U^\dagger(-t, t_0)$.

Ok the stuff follows by Taylor.

Lecture 5: Solving for unitary operators

Thu 18 Sep 2025 14:03

We are solving for

$$\frac{\partial U}{\partial t} = \frac{1}{i\hbar} H U, \quad U(t_0, t_0) = 1.$$

Of course, recall that

$$U(t, t_0) = \exp\left(-\frac{i}{\hbar} H(t)(t - t_0)\right)$$

does not work since $[H(t), H(t')] \neq 0$ in general. For time independent hamiltonians, it works I think.

Before solving this, we shift to the Dirac interaction picture. Time dependent hamiltonians are generally given by $H(t) = H_0 + V(t)$, where H_0 is time independent, and V is the *interaction term*. We will define

$$|\Psi_I(t)\rangle := \exp\left(\frac{i}{\hbar} H_0(t - t_0)\right) |\Psi(t)\rangle.$$

The inverse of the operator is given by multiplying through the exponent by -1 . This new ket is called the Dirac interaction ket. They agree on t_0 . We then want a time evolution in this picture

$$|\Psi_I(t)\rangle = U_I(t, t_0) |\Psi_I(t_0)\rangle.$$

We want an equation of motion for U_I . Since they agree on the initial state,

$$|\Psi_I(t)\rangle = \exp\left(\frac{iH_0}{\hbar}(t - t_0)\right) U(t, t_0) |\Psi_I(t_0)\rangle.$$

We thus necessarily have that the exponential times U is U_I . Then

$$\begin{aligned} \frac{\partial U_I}{\partial t} &= \frac{i}{\hbar} H_0 \exp\left(\frac{i}{\hbar} H_0(t - t_0)\right) U + \exp\left(\frac{i}{\hbar} H_0(t - t_0)\right) \frac{\partial U}{\partial t} = \frac{i}{\hbar} H_0 \exp\left(\frac{i}{\hbar} H_0(t - t_0)\right) U + \exp\left(\frac{i}{\hbar} H_0(t - t_0)\right) \frac{1}{i\hbar} H U \\ &= \frac{i}{\hbar} H_0(t - t_0) U + \exp\left(\frac{i}{\hbar} H_0(t - t_0)\right) \frac{-i}{\hbar} (H_0 + V) U. \end{aligned}$$

Substitute in our expression for U in terms of U_I :

$$\begin{aligned} &= \frac{i}{\hbar} \exp\left(\frac{i}{\hbar} H_0(t - t_0)\right) \exp\left(-\frac{i}{\hbar} H_0(t - t_0)\right) U_I + \frac{i}{\hbar} \exp\left(\frac{i}{\hbar} H_0(t - t_0)\right) (-H_0 - V) \exp\left(-\frac{i}{\hbar} H_0(t - t_0)\right) U_I \\ &= \frac{i}{\hbar} H_0 U_I + \frac{i}{\hbar} (-H_0) U_I - \frac{i}{\hbar} \exp\left(\frac{i}{\hbar} H_0(t - t_0)\right) V \exp\left(-\frac{i}{\hbar} H_0(t - t_0)\right) U_I. \end{aligned}$$

I probably wrote this down a bit wrong. The first two terms cancel, giving

$$\frac{\partial U_I}{\partial t} = -\frac{i}{\hbar} \exp\left(\frac{i}{\hbar} H_0(t - t_0)\right) V(t) \exp\left(-\frac{i}{\hbar} H_0(t - t_0)\right) U_I.$$

We define the exponential time $V(t)$ to be $V_I(t)$, and call it the *potential*. In this form,

$$\frac{\partial U_I}{\partial t} = \frac{1}{i\hbar} V_I(t) U_I(t, t_0).$$

We solve this given an initial condition $U(t_0, t_0) = 1$. This is the Dirac interaction picture, and is extremely important.

To solve this, we pass to an integral equation

$$(U_I = U_I(t) - 1) = \int_{t_0}^t \frac{1}{i\hbar} V_I U_I dt.$$

This is solved by Picard iterations. I don't want to work it all out you get $U_I^{(0)}(t, t_0) = 1$, and inductively

$$U_I^{(n)}(t, t_0) = 1 + \frac{1}{i\hbar} \int_{t_0}^t V_I(t) U_I^{(n-1)}(t, t_0) dt.$$

We then take $n \rightarrow \infty$. This gives something like

$$U_I(t, t_0) = 1 + \sum_{i=1}^{\infty} \left(\frac{1}{i\hbar} \right)^i \int_{t_0}^t \prod_{j=1}^i V_I(t^j) dt.$$

These are Dyson series. These play a fundamental role in the path integral formulation, which will be done in 512. This is also important in relativistic models.

Lecture 6: Nuclear magnetic resonance

Tue 30 Sep 2025 13:59

Recall from my notes that the most general time-independent hamiltonian for a 2-dimensional state is

$$H = \begin{pmatrix} \varepsilon_0 & \gamma e^{-i\chi} \\ \gamma e^{i\chi} & \varepsilon_1 \end{pmatrix}.$$

Then

$$P_{0 \rightarrow 1}(t) = \frac{4\gamma^2}{4\gamma^2 + (\varepsilon_1 - \varepsilon_0)^2} \sin^2 \left(\frac{t}{2\hbar} \sqrt{4\gamma^2 + (\varepsilon_1 - \varepsilon_0)^2} \right).$$

A spin 1/2 proton in a uniform $\mathbf{B} + R\mathbf{F}$ field has

$$H(t) = -\boldsymbol{\mu} \cdot \mathbf{B}(t) = -\boldsymbol{\mu} \cdot (B_0 \mathbf{z} + \mathbf{B}_1(t)).$$

We want to find the transmission probability $P_{0 \rightarrow 1}$. Here, $|0\rangle$ is the ground state. B_1 is small compared to B_0 , so we would expect that the transmission probability should be small ($|0\rangle$ eigenstate for ONLY B_0 , so small contribution does small change).

We could discuss a linearly polarized field $\mathbf{B}_1 = B_1 \mathbf{x} \cos \omega t$. We could also discuss a circular polarized field

$$B_0 \mathbf{z} + B_1 (\mathbf{x} \cos \omega t \pm \mathbf{y} \sin \omega t).$$

So then if you take the dot product for the $-$ case of the circular case, then

$$\mathbf{H} = -\frac{1}{2}\hbar = \omega_L \sigma_z - \frac{1}{2}\hbar \Omega_1 (\sigma_x \cos \omega t - \sigma_y \sin \omega t).$$

Fascinatingly, there *are not exact solutions* to the linear case. There *are* exact solutions to the circularly polarized case, which we explore.

We will rewrite the hamiltonian as

$$\mathbf{H} = \begin{pmatrix} -\frac{1}{2}\hbar \omega_L & \gamma e^{i\omega t} \\ \gamma e^{-i\omega t} & \frac{1}{2}\hbar \omega_L \end{pmatrix}.$$

Here, $\gamma = -\frac{1}{2}\hbar\Omega_1$. We can rewrite this as

$$\mathbf{H}_0 + \mathbf{V}(t).$$

We have

$$\mathbf{H}_0 = \text{diag} \left(-\frac{1}{2}\hbar\omega_L, \frac{1}{2}\hbar\omega_L \right), \quad \mathbf{V}(t) = \begin{pmatrix} & \gamma e^{i\omega t} \\ \gamma e^{-i\omega t} & \end{pmatrix}.$$

We could solve this with the Schrödinger picture by using $i\hbar\partial_t|\Psi\rangle = \mathbf{H}|\Psi\rangle$. We would then set $|\Psi\rangle = \alpha_0(t)|0\rangle + \alpha_1(t)|1\rangle$, and solve the differential equation. However, it is better to solve this in Dirac's interaction picture. This will be a homework problem. Our reference frame changes to

$$|\Psi_I\rangle = \exp \left(\frac{i}{\hbar} \mathbf{H}_0 t \right) |\Psi_0\rangle.$$

This satisfies the equation of motion

$$i\hbar \frac{\partial}{\partial t} |\Psi_I\rangle = \exp \left(\frac{i}{\hbar} \mathbf{H}_0 t \right) \mathbf{V}(t) \exp \left(-\frac{i}{\hbar} \mathbf{H}_0 t \right) |\Psi_0\rangle.$$

We then just solve this easier equation.

There is yet another frame, the Rabi rotating frame. Here, we jump onto the $\mathbf{B}_1$ frame. We write

$$|\Psi_R\rangle = \exp \left(-\frac{i\omega t}{2} \sigma_z \right) |\Psi\rangle \iff |\Psi\rangle = \exp \left(\frac{i\omega t}{2} \sigma_z \right) |\Psi_R\rangle.$$

We would like to find an equation

$$i\hbar \frac{\partial}{\partial t} |\Psi_R\rangle = \mathbf{H}_R |\Psi_R\rangle.$$

We will then attain a definition of our hamiltonian. We substitute the Schrödinger picture:

$$i\hbar \frac{\partial}{\partial t} \exp \left(-\frac{i\omega t}{2} \sigma_z \right) |\Psi\rangle = i\hbar \left(-\frac{i\omega}{2} \sigma_z \right) \exp \left(-\frac{i\omega t}{2} \right) |\Psi\rangle + \exp \left(-\frac{i\omega t}{2} \sigma_z \right) i\hbar \frac{\partial}{\partial t} |\Psi\rangle.$$

Applying the Schrödinger equation:

$$= i\hbar \left(-\frac{i\omega}{2} \sigma_z \right) \exp \left(-\frac{i\omega t}{2} \right) |\Psi\rangle + \exp \left(-\frac{i\omega t}{2} \sigma_z \right) (\mathbf{H}_0 + \mathbf{V}(t)) |\Psi\rangle.$$

Plugging in Ψ_R for Ψ gives a mess

$$= \frac{\hbar\omega}{2} \sigma_z + \exp \left(-\frac{i\omega t}{2} \sigma_z \right) \mathbf{H}_0 \exp \left(\frac{i\omega t}{2} \sigma_z \right) + \exp \left(-\frac{i\omega t}{2} \sigma_z \right) \mathbf{V}(t) \exp \left(\frac{i\omega t}{2} \sigma_z \right) |\Psi_R\rangle.$$

Because $\mathbf{H}_0 = -\frac{1}{2}\hbar\omega_L\sigma_z$ is a scalar multiple of σ_z , the exponentials commute with $\mathbf{H}_0$ and cancel:

$$\mathbf{H}_R = \frac{\hbar}{2} (\omega - \omega_L) \sigma_z + \exp \left(-\frac{i\omega t}{2} \sigma_z \right) \mathbf{V}(t) \exp \left(\frac{i\omega t}{2} \sigma_z \right) |\Psi_R\rangle.$$

This is the Hamiltonian in the Rabi frame. Can we compute the $\mathbf{V}_R(t)$ term, which is the $\mathbf{V}$ term? Recall that

$$e^{-i\omega t\sigma_z/2} = \begin{pmatrix} e^{-i\omega t/2} & \\ & e^{i\omega t/2} \end{pmatrix}.$$

This gives

$$\mathbf{V}_R(t) = \begin{pmatrix} e^{-i\omega t/2} & \\ & e^{i\omega t/2} \end{pmatrix} \begin{pmatrix} & \gamma e^{i\omega t} \\ e^{-i\omega t} & \end{pmatrix} \begin{pmatrix} e^{i\omega t/2} & \\ & e^{-i\omega t/2} \end{pmatrix}.$$

We can check that this becomes

$$\mathbf{V}_R(t) = \gamma \sigma_x.$$

Therefore,

$$\mathbf{H}_R = \begin{pmatrix} -\frac{\hbar}{2}(\omega_L - \omega) & \gamma \\ \gamma & \frac{\hbar}{2}(\omega_L - \omega) \end{pmatrix}.$$

This is time independent, so it is given by the solution I did on Friday. The ground state is the top left term, and the excited state is the bottom right term, so plugging this into the formula at the top of this lecture gives

$$P_{0 \rightarrow 1}(t) = \frac{\gamma^2/\hbar^2}{\frac{\gamma^2}{\hbar^2} + \frac{(\omega - \omega_L)^2}{4}} \sin^2 \left(t \sqrt{\frac{\gamma^2}{\hbar^2} + \frac{(\omega_L - \omega)^2}{4}} \right).$$

We have written it in a slightly different form here. This is the generalized Rabi frequency.

Why did this frame turn the Hamiltonian into a time-independent Hamiltonian? Since the field is circularly polarized, jumping on the circular rotation yields a fixed $\mathbf{B}$ -field, so it is time independent. When $\omega = \omega_L$, we are at a resonant state, so $\omega - \omega_L$ is called the *detuning*.

At resonance, the Rabi flopping frequency oscillates from 0 to 1. All we have in the Rabi frame is a static field, so it just precesses around it. So every half period of a rotation, it has flipped eigenstates.

We use this to construct quantum gates. For example, σ_x is given by turning on a field to allow it to precess at resonance by π , and then turning it off. We can actually construct *any* gate like this apparently.

Lecture 7: Gauge invariance

Thu 30 Oct 2025 14:01

So in QM a gauge transform is

$$|\Psi\rangle \rightarrow |\tilde{\Psi}\rangle = \exp\left(\frac{iv}{\hbar}\Lambda(\mathbf{x}, t)\right) |\Psi\rangle.$$

So in classical E&M, the Gauge potential A is a vector quantity satisfying $\mathbf{B} = \nabla \times A$. I write non-bold A because typing Abf inputs $\mathcal{A}bf$. Also

$$\mathbf{E} = -\nabla\Phi - \frac{\partial A}{\partial t}.$$

The most fundamental law of E&M is conservation of charge

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.$$

Gauge freedom says that physics does not change under transformations

$$A \rightarrow A + \nabla\Lambda$$

or

$$\Phi \rightarrow \Phi - \frac{\partial \Lambda}{\partial t}$$

for some scalar Λ . Only potentials that can be connected by a scalar are valid. For some reason we write

$$\mathbf{E} = \frac{-\mu_0 c^2}{4\pi} \frac{e}{r^2} \hat{r}$$

instead of

$$-\frac{1}{4\pi\epsilon_0} \frac{e}{r^2} \hat{r}$$

for a point charge of charge e . If we do a very long discussion then we find the *relativistic* energy for a charged particle in an EM field is given by swapping $\vec{p}$ for $\vec{p} - qA$. Here $\vec{p} = -i\hbar\nabla$ and A is a Gauge transformation or something. SO this is the relativistic momentum operator or something.